

Logistic Mixed Models

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1 Using `glmer`

To fit the GLMM we're going to use `glmer` in the `lme4` package. We previously used the `lme4` package to fit linear mixed models with `lmer`.

The arguments to `glmer` are:

- **formula**: a two-sided linear formula. Random-effects terms are distinguished by vertical bars (“|”) separating expressions for design matrices from grouping factors.
- **data**: an optional data frame.
- **family**: a GLM family, see `glm` and `family`.
- **nAGQ**: integer scalar - the number of points per axis for evaluating the adaptive Gauss-Hermite approximation to the log-likelihood. Defaults to 1, corresponding to the Laplace approximation. Values greater than 1 produce greater accuracy in the evaluation of the log-likelihood at the expense of speed. A value of zero uses a faster but less exact form of parameter estimation for GLMMs by optimizing the random effects and the fixed-effects coefficients in the penalized iteratively reweighted least squares step. (See Details.)

In our models below, the computation time is pretty minimal. As a result, we're going to set `nAGQ=20` which is a relatively large value.

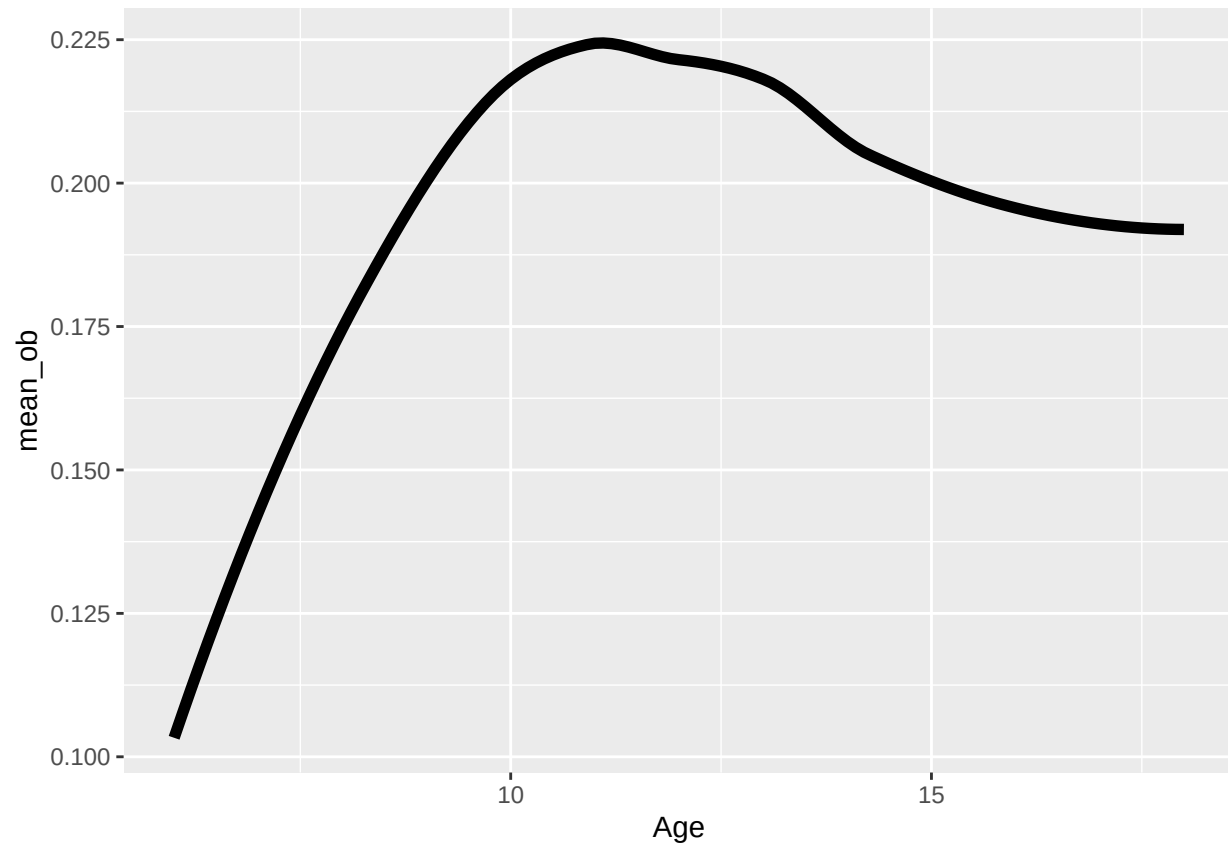
2 Muscatine Coronary Risk Factor Study

- The Muscatine Coronary Risk Factor Study was a study designed to look at the change in obesity in children.
- Five age cohorts were measured in 1977, 1979 and 1981.
- The study had had five cohorts of children, initially aged 5–7, 7–9, 9–11, 11–13, and 13–15.
- Children were classified as obese or not obese.

Let's look at the trends with age by Obesity.

```
# First, get the trend of Age by Obesity
MCRS_loess <- loess(OB_ind ~ Age_k, data = MCRS_longest)
# Second, data for Age and predicted Obesity in a useable form.
MCRS_summary <- tibble(
  Age = seq(6, 18, 0.1),
  mean_ob = predict(MCRS_loess, data.frame(Age_k = seq(6, 18, 0.1))))
```

```
# Third, plot the age and predicted obesity trends.
p <- ggplot(MCRS_summary, aes(x = Age, y = mean_ob) ) +
  geom_line(linewidth = 2)
p
```

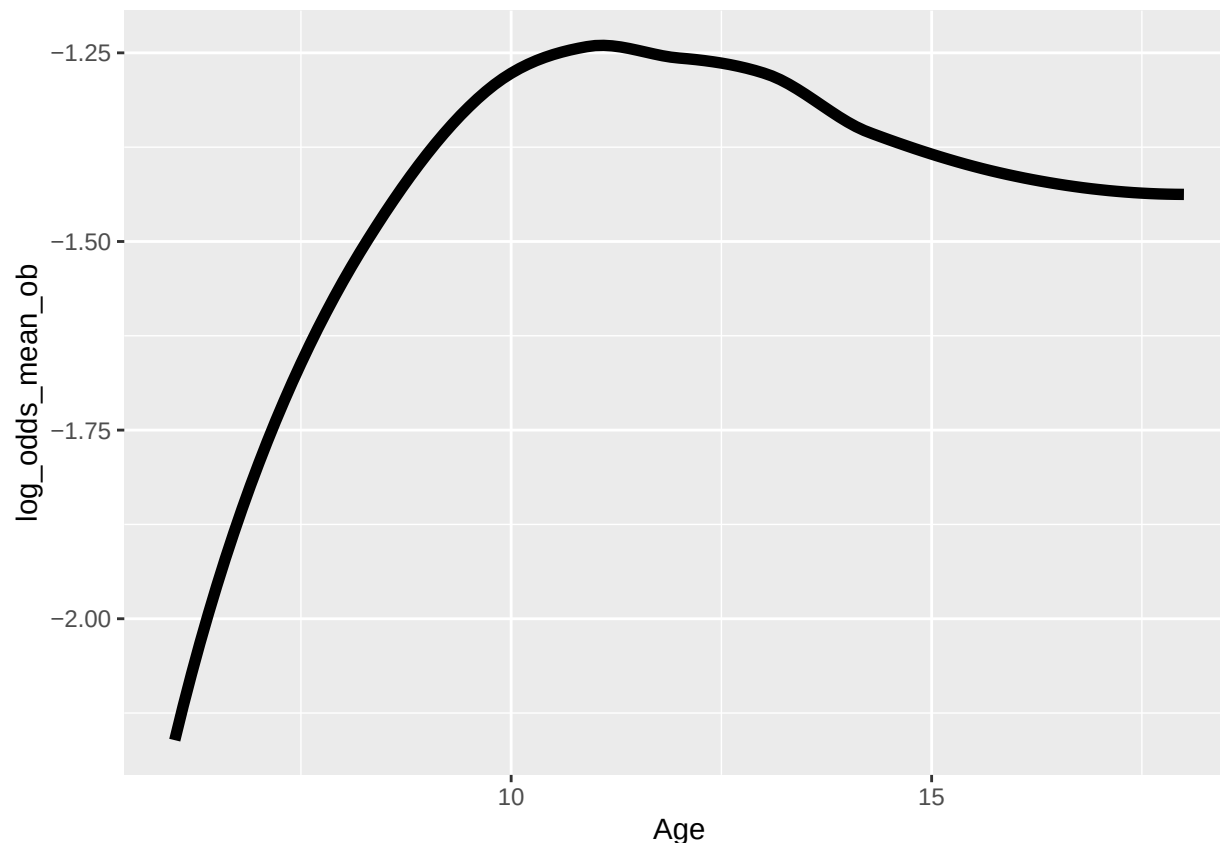


However, recall that we model

$$\log \left\{ \frac{P(OB = 1)}{1 - P(OB = 1)} \right\} = \text{predictors}$$

As a result, we don't want to see the relationship between `mean_ob` and `Age`, we want to see the trend in the log odds of `mean_ob` and `Age`. So, let's look at that.

```
MCRS_summary <- MCRS_summary %>%
  mutate(log_odds_mean_ob = log(mean_ob/(1-mean_ob)))
ggplot(MCRS_summary, aes(x = Age, y = log_odds_mean_ob) ) +
  geom_line(linewidth = 2)
```



We can look at this using a linear and log trends for age.

```
library(lme4)
library(lmerTest)

form <- OB_ind ~ Age_k + log(Age_k) + factor(Cohort) + (1|ID)
system.time(log_mod <- glmer(formula = form, data = MCRS_longest, nAGQ = 10, family = binomial))

##      user  system elapsed
## 24.194   2.342  26.629

summary(log_mod)

## Generalized linear mixed model fit by maximum likelihood (Adaptive
## Gauss-Hermite Quadrature, nAGQ = 10) [glmerMod]
## Family: binomial ( logit )
## Formula: OB_ind ~ Age_k + log(Age_k) + factor(Cohort) + (1 | ID)
## Data: MCRS_longest
##
##      AIC      BIC   logLik deviance df.resid
## 3509.1   3561.8 -1746.6  3493.1     5302
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.22765 -0.38534 -0.01076 -0.00864  3.10208
##
## Random effects:
## Groups Name      Variance Std.Dev.
```

```

## ID      (Intercept) 47.72    6.908
## Number of obs: 5310, groups:  ID, 154
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -17.0400    1.9611  -8.689 < 2e-16 ***
## Age_k         -0.4107    0.1081  -3.798 0.000146 ***
## log(Age_k)     5.2401    1.2067   4.342 1.41e-05 ***
## factor(Cohort)2  0.2963    0.1641   1.806 0.070924 .
## factor(Cohort)3  0.2856    0.1876   1.522 0.127916
## factor(Cohort)4 -0.1066    0.2129  -0.501 0.616531
## factor(Cohort)5 -0.0396    0.2464  -0.161 0.872313
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) Age_k  lg(A_) fc(C)2 fc(C)3 fc(C)4
## Age_k         0.802
## log(Age_k)   -0.847 -0.974
## fcctr(Chrt)2  0.245  0.257 -0.322
## fcctr(Chrt)3  0.304  0.240 -0.357  0.742
## fcctr(Chrt)4  0.214  0.066 -0.218  0.679  0.776
## fcctr(Chrt)5  0.048 -0.169 -0.001  0.575  0.696  0.787

```